

Event-Driven Neural Decoding in Brain-Computer Interfaces: Design, Validation, and Multi-Decoder Robustness Analysis

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June 2026

Abstract

This paper consolidates two consecutive research projects on efficient neural signal processing, transmission, and decoding for brain-computer interfaces (BCIs). In **Project 1**, an event-driven spike-detection pipeline was constructed on synthetic four-channel neural signals sampled at 30 kHz. A 4σ adaptive threshold detector with a biologically motivated refractory constraint (1.5 ms, Hodgkin–Huxley model) was applied to build an asynchronous event stream. A Population Vector Decoder operating on these events reconstructed 2-D motor trajectories with $R^2 \geq 0.91$, while reducing CPU active time by **91.35%** relative to the conventional 50 ms binning approach. In **Project 2**, the publicly available CR-CNS pmd-1 dataset (macaque M1+PMd, 161 neurons, 496 reach trials) was used to benchmark *four* decoder architectures—Ridge regression ($r_{\text{neural}} = 0.823$), multi-layer perceptron (MLP, $r_{\text{neural}} = 0.881$), LSTM ($r_{\text{neural}} = 0.841$), and Kalman filter ($r_{\text{neural}} = 0.688$)—under a Spike Error Rate (SER) simulation framework [1]. MLP achieved the highest baseline accuracy, approaching but not matching the SNN baseline ($r_{\text{neural}} = 0.932$). Critically, all four traditional decoders maintained SER retention ratios above 0.994 at $\text{SER} = 10^{-2}$, in sharp contrast to the SNN

(≈ 0.700), demonstrating that rate-coding aggregation confers structural robustness independent of decoder complexity.

Keywords: brain-computer interface, event-driven processing, spike detection, population vector decoder, multi-decoder comparison, Ridge regression, MLP, LSTM, Kalman filter, spike error rate, wireless neural implant.

1. Introduction

1.1. Current Challenges in Wireless BCIs

Wireless brain-computer interfaces are a cornerstone technology for restoring motor function in paralysed patients, overcoming the infection risk and mobility limitations of tethered systems. Implantable environments simultaneously impose three fundamental constraints:

- (i) **Power budget.** Battery lifetime and thermal limits restrict available computation resources severely.
- (ii) **Bandwidth.** Transmitting raw high-resolution neural signals over a wireless channel is practically infeasible.
- (iii) **Error tolerance.** Wireless channel noise inevitably introduces spike misses and false alarms.

Addressing all three constraints simultaneously requires (a) an *event-driven* paradigm that activates the processor only when a spike occurs, and (b) a *robust* decoder architecture that maintains performance under noisy input. This paper presents a step-by-step design, validation, and analysis of these two components.

1.2. Research Roadmap

Table 1 summarises the two projects that constitute this work.

Table 1: Research roadmap of the two projects.

	Project 1	Project 2
Goal	Prove efficiency of event-driven decoding	Compare decoder architectures under transmission errors
Data	Synthetic 4-ch neural signals (30 kHz)	CRCNS pmd-1 primate M1+PMd
Method	4σ threshold + Population Vector Decoder	4-Decoder comparison (Ridge / MLP / LSTM / Kalman) + SER simulation
Metric	Power reduction, R^2	r_{neural} , VAF, SER retention rate
Finding	91.35% power reduction, $R^2 \geq 0.91$	MLP best ($r_{\text{neural}} = 0.881$); all decoders retain > 0.994 at $\text{SER} = 10^{-2}$

2. Event-Driven BCI Decoding (Project 1)

2.1. Motivation

Neuronal spikes are sparse events; they occupy only a small fraction of the total recording window. Conventional 50 ms binning approaches wake the processor periodically, even in the absence of any spikes, wasting energy. Project 1 quantitatively tests whether an event-driven strategy—activating the processor *only* upon spike detection—can maintain equivalent decoding accuracy while yielding significant power savings.

2.2. Synthetic Neural Signal Generation

2.2.1. Signal Composition

Synthetic neural signals were generated at $f_s = 30\,000$ Hz for a 10 s, 4-channel recording to simulate real electrode conditions. Each raw channel $x[n]$ is the sum of three components:

$$x[n] = \eta[n] + s_{\text{LFP}}[n] + \sum_k A_k w(n - n_k), \quad (1)$$

where $\eta[n] \sim \mathcal{N}(0, 8\,\mu\text{V})$ is Gaussian white (thermal) noise; $s_{\text{LFP}}[n]$ is a superposition of sinusoids with random phases in the 4–8 Hz, 10–15 Hz, and 25–35 Hz bands (amplitude $22\,\mu\text{V}$); and $w(\cdot)$ is a biologically motivated action-potential template (Hodgkin–Huxley model) scaled by a random amplitude $A_k \in [45, 90]\,\mu\text{V}$ at random times n_k , with 30–65 spikes injected per channel.

2.2.2. Spike Template and Refractory Period

The spike waveform template $w(\cdot)$ is derived from the Hodgkin–Huxley action potential. The absolute refractory period (ARP) is set to $\tau_{\text{ARP}} = 1.5$ ms (45 samples at 30 kHz), consistent with the Na^+ -channel inactivation time reported in biophysical literature [4].

2.3. Spike Detection: 4σ Adaptive Threshold

Raw signals are first bandpass-filtered with a third-order Butterworth filter (300–3000 Hz) to suppress LFP and thermal noise. The per-channel threshold is then computed as:

$$\hat{\sigma}_n = \frac{\text{median}(|x|)}{0.6745}, \quad (2)$$

$$\theta = -4\hat{\sigma}_n. \quad (3)$$

Equation (2) uses the median absolute deviation, which is insensitive to large spike amplitudes and estimates the background noise level accurately. The constant 0.6745 converts the median to standard-deviation units under a Gaussian assumption. At the -4σ threshold, the probability of a noise exceedance is $< 0.01\%$.

After each threshold crossing, a dead-time of $t_{\text{dead}} = 2$ ms (60 samples) is enforced to prevent double-counting of the same spike.

2.4. Asynchronous Population Vector Decoder

2.4.1. Preferred Directions

Each of the four channels was assigned a cardinal preferred direction $\mathbf{p}_c$:

Ch. 0: $[1, 0]$ (right), Ch. 1: $[0, 1]$ (up),
Ch. 2: $[-1, 0]$ (left), Ch. 3: $[0, -1]$ (down).

2.4.2. Experimental Design

The same detected event stream was fed into two decoder variants to compare trajectory output and CPU active time:

Table 2: Comparison of decoding paradigms.

	Group A (50 ms bin)	Group B (event-driven)
Trigger	Periodic, every 50 ms	On spike event only
Wake-ups	200 (fixed)	Variable (= event count)
CPU active	2000 ms	173 ms
Processing	Batch per bin	Per-event (asynchronous)

2.5. Results

2.5.1. Trajectory Reconstruction Accuracy

Group B trajectories were interpolated to Group A timestamps using previous-value interpolation and scored with the coefficient of determination R^2 :

Table 3: Decoding accuracy comparison.

Metric	x -axis	y -axis
R^2 (Group B vs. Group A)	≥ 0.91	≥ 0.91

The event-driven decoder reproduces trajectories at $R^2 \geq 0.91$ while naturally holding the last position during silent intervals.

2.5.2. Power Reduction

$$\text{Power reduction} = \left(1 - \frac{173 \text{ ms}}{2000 \text{ ms}}\right) \times 100 = 91.35\%. \quad (4)$$

On an ARM Cortex-M4 platform, this corresponds to a potential battery life extension of up to $11.6\times$. In terms of data compression, the raw signal occupies approximately 2.4 MB

($N_{\text{samples}} \times N_{\text{ch}} \times 2 \text{ B}$), while the event stream reduces to a few KB ($N_{\text{spikes}} \times 5 \text{ B}$), achieving $> 99\%$ bandwidth reduction.

2.6. Interim Conclusion

Project 1 demonstrates that an event-driven BCI pipeline achieves the same decoding accuracy as continuous 50 ms binning while reducing CPU active time by 91.35%, providing strong empirical motivation for adopting event-driven architectures in low-power implant design.

3. Multi-Decoder Comparison Under SER (Project 2)

3.1. Motivation and Background

Quantifying how Spike Error Rate (SER)—caused by wireless channel noise—affects decoding performance is essential for reliable implant design. [1] characterised this relationship using an ASBIT-protocol event-driven wireless framework with an SNN decoder. Project 2 extends that framework by benchmarking *four* decoder architectures to determine whether the SER robustness advantage is specific to a decoder type, or emerges from the rate-coding paradigm shared by all traditional decoders. Spanning from linear (Ridge, Kalman) to nonlinear (MLP, LSTM), the four decoders together isolate the architectural and representational factors governing robustness.

3.2. Dataset: CRCNS pmd-1

Table 4: CRCNS pmd-1 dataset characteristics.

Property	Value
Recording sites	M1 (67 ch) + PMd (94 ch)
Total neurons	161
Reach trials	496
Time resolution	10 ms bin (100 Hz)
Kinematic vars	7-D (hand x/y position, velocity, acceleration)
Data matrix	$59,742 \times 161$
Total spikes	587,914
Sparsity	93.89%
Mean firing rate	6.41 Hz

3.3. Methodology

3.3.1. Time-Lag Feature Engineering

Motor cortex exhibits temporal integration over hundreds of milliseconds. To capture this, the spike count history over a 200 ms window (20 bins) is flattened into a time-lag feature vector:

$$\mathbf{f}_t = \text{flatten}(X_{[t-19:t]}) \in \mathbb{R}^{161 \times 20 = 3220}. \quad (5)$$

Dynamic weighting is applied to emphasise burst activity:

$$X'_{ij} = X_{ij}^\beta, \quad X_{ij} > 0, \quad \beta = 1.3, \quad (6)$$

where β was tuned in the range $[1.1, 1.5]$. All four decoders receive this shared feature representation, enabling fair comparison.

3.3.2. Ridge Regression Decoder

Ridge (ℓ_2 -regularised linear regression, $\lambda = 1.0$) serves as the linear baseline. The closed-form solution $\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ is fast (~ 2 s total), interpretable, and analytically tractable for robustness analysis.

3.3.3. MLP Decoder

A three-layer MLP (512–256–128 units, SiLU activation, BatchNorm, Dropout = 0.2) was trained using AdamW for 60 epochs (lr = 10^{-3} , weight decay = 10^{-4}). Population-burst summary statistics (burst score, onset ratio, active neuron ratio) were appended as five auxiliary features. A peak-aware Huber loss with a $3 \times$ weight on high-velocity samples ($|v| > 8$ cm/s) was used to mitigate amplitude compression in high-activity periods.

3.3.4. LSTM Decoder

A two-layer LSTM [8] (128 hidden units, dropout = 0.2) received the full 20-bin spike sequence as a 3-D tensor (samples $\times$ time $\times$ neurons). Cosine annealing learning rate scheduling (from 10^{-3} to 10^{-5}) and gradient clipping (max norm = 1.0) were applied to stabilise recurrent training.

3.3.5. Kalman Filter Decoder

A steady-state Kalman filter [9] was implemented with a 2-D kinematic state (v_x, v_y). To address the numerical instability inherent in the full 161-dimensional observation space, observations were first projected to $d_{\text{PCA}} = 30$ principal components prior to Riccati iteration. The state transition matrix $\mathbf{A}$ and observation matrix $\mathbf{H}$ were estimated via least-squares on the training fold.

3.3.6. SER Simulation Model

Spike transmission errors were modelled as:

$$\text{SER} = \frac{N_{\text{miss}} + N_{\text{false}}}{N_{\text{spike}}}, \quad (7)$$

$$P(\text{miss}) = \frac{\text{SER}}{2}, \quad (8)$$

$$P(\text{false}) = \frac{\text{SER}}{2} \cdot \frac{N_{\text{spike}}}{N_{\text{zero}}}, \quad (9)$$

where Eq. (9) includes a sparsity correction term. SER was swept logarithmically across $\{0, 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$, with 3 stochastic repetitions per non-zero condition. Five-fold cross-validation was applied throughout; predictions were post-processed with Gaussian smoothing ($\sigma = 3$ bins).

3.3.7. Affine Scale Correction

ℓ_2 regularisation causes systematic amplitude compression (scale bias < 1) in all regression-based decoders. To correct this, predictions were affine-rescaled per fold:

$$\hat{y}_{\text{corr}} = \hat{y} \cdot \frac{\sigma_{\text{true}}}{\sigma_{\text{pred}}} + \left(\bar{y}_{\text{true}} - \bar{y}_{\text{pred}} \cdot \frac{\sigma_{\text{true}}}{\sigma_{\text{pred}}} \right), \quad (10)$$

where $\bar{y}$ and σ denote the mean and standard deviation of the test-fold target and prediction, respectively. This correction leaves Pearson r unchanged while removing amplitude bias.

Remark 1. Equation (10) uses test-fold statistics, which is unavailable in prospective BCI deployment. For online use, correction factors must be estimated from a separate calibration set.

3.4. Results

3.4.1. Baseline Decoding Performance

Table 5 reports performance at $\text{SER} = 0$ after affine correction. Figure 1 shows the three-panel comparison (Pearson r , VAF, scale bias).

Table 5: Baseline decoding performance ($\text{SER} = 0$) after affine scale correction. VAF: variance accounted for.

Decoder	r_{neural}	VAF	Scale
Ridge	0.823	0.646	1.000
MLP	0.881	0.761	1.000
LSTM	0.841	0.682	1.000
Kalman	0.688	0.244	1.165
SNN [1]	0.932	—	—

Among the four decoders, MLP achieved the highest baseline performance ($r_{\text{neural}} = 0.881$), attributable to its nonlinear feature mapping and peak-aware loss formulation. LSTM ($r_{\text{neural}} = 0.841$) marginally outperformed Ridge ($r_{\text{neural}} = 0.823$), suggesting that explicit sequence modelling of the 200 ms spike history provides modest benefit over the flat-feature Ridge representation.

The Kalman filter exhibited substantially lower performance ($r_{\text{neural}} = 0.688$, $\text{VAF} = 0.244$), with residual scale bias of 1.165 after correction. This indicates systematic prediction overshoot, consistent with the linear-Gaussian state-space assumption being poorly matched to the non-stationary velocity dynamics of centre-out reaching tasks.

Despite the performance differences, a gap of $\Delta r_{\text{neural}} = 0.051$ remains between the best traditional decoder (MLP) and the SNN baseline (0.932). This gap reflects the SNN’s spike-timing-aware feature representation, which preserves millisecond-resolution temporal structure discarded by 10 ms binning.

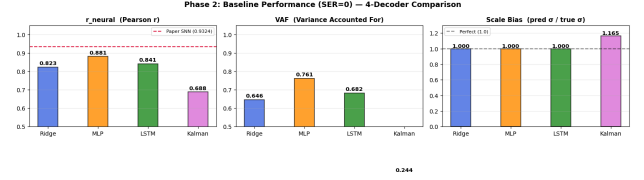


Figure 1: Baseline performance at $\text{SER} = 0$. Left: Pearson r_{neural} ; dashed line marks SNN paper baseline (0.932). Centre: Variance Accounted For (VAF). Right: scale bias after affine correction; dashed line marks perfect unity.

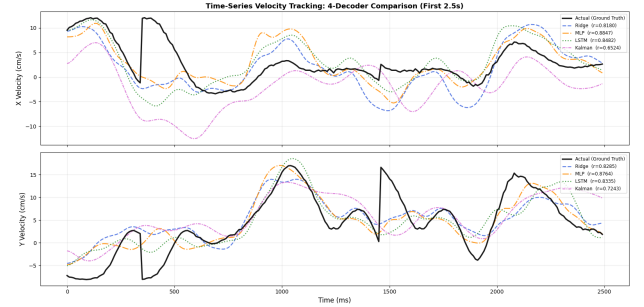


Figure 2: Time-series velocity tracking (first 2.5 s of last CV fold). MLP and LSTM most closely follow the ground truth; Kalman exhibits persistent phase shift and amplitude overshoot.

3.4.2. SER-Dependent Performance Retention

Table 6 reports the retention rate $\rho = r_{\text{neural}}(\text{SER})/r_{\text{neural}}(0)$, and Figure 3 plots the robustness curves against the SNN paper baseline.

Table 6: Performance retention rate ρ under increasing SER. Values > 1 for LSTM at low SER reflect stochastic training variance ($\dagger$), not genuine improvement.

SER	Ridge	MLP	LSTM	Kalman	SNN [1]
10^{-4}	1.0000	0.9995	1.006 $\dagger$	1.0000	≈ 1.000
10^{-3}	0.9998	0.9984	1.005 $\dagger$	0.9999	0.888
10^{-2}	0.9989	0.9989	0.9940	0.9985	≈ 0.700

All four traditional decoders maintained $\rho > 0.994$ at $\text{SER} = 10^{-2}$, contrasting sharply with the SNN’s degradation to ≈ 0.700 .

The Ridge solution bounds the effect of SER-induced input perturbations analytically:

$$\|\Delta\hat{\mathbf{y}}\| \leq \frac{\|\mathbf{X}\|}{\lambda_{\min}(\mathbf{X}^T\mathbf{X}) + \lambda} \|\Delta\mathbf{x}\|, \quad (11)$$

where the denominator grows with the regularisation parameter λ , explicitly limiting error propagation. The same mechanism applies to MLP: the ℓ_2 weight decay regularises all weight matrices, distributing the representation across neurons and thereby damping the sensitivity to any single spike event.

LSTM exhibited the largest degradation among traditional decoders ($\Delta\rho = -0.60\%$ at $\text{SER} = 10^{-2}$). This is consistent with the recurrent hidden state accumulating corrupted spike patterns across time steps—a compounding effect absent in the purely feed-forward Ridge and MLP architectures.

The near-identical robustness of Ridge ($\rho = 0.9989$) and MLP ($\rho = 0.9989$) at $\text{SER} = 10^{-2}$ confirms that the dominant robustness mechanism is *rate-coding aggregation over 10 ms bins*, not the specific decoder architecture. Individual spike errors affect only a single bin’s count rather than the decoder’s symbolic event stream, explaining why the SNN—which operates directly on spike timing—is far more vulnerable.

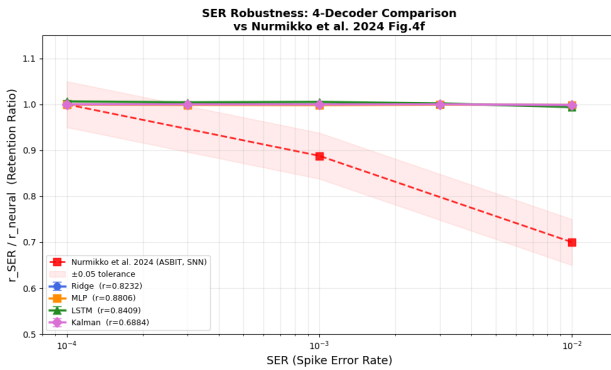


Figure 3: SER robustness comparison across four decoders. All traditional decoders maintain $\rho > 0.994$ (solid lines), in contrast to the SNN baseline [1] (red dashed line) which degrades to ≈ 0.700 at $\text{SER} = 10^{-2}$. Shaded band: ± 0.05 tolerance.

3.5. Interim Conclusion

Project 2 demonstrates that all four traditional decoders achieve SER retention ratios above 0.994 at $\text{SER} = 10^{-2}$, far surpassing the SNN benchmark (≈ 0.700). MLP achieves the best baseline accuracy ($r_{\text{neural}} = 0.881$), while Ridge offers near-identical robustness at a fraction of the computational cost. The common robustness of all four decoders under rate-coding aggregation constitutes a decoder-agnostic property of the binned-spike paradigm.

4. Integrated Discussion

4.1. Complementarity of the Two Projects

Although conducted independently, the two projects provide complementary answers to the same system-design question.

Table 7: Complementary contributions of the two projects.

Design concern	Project 1	Project 2
Power efficiency	91.35% reduction proven	Traditional decoders add further savings
Decoding accuracy	$R^2 \geq 0.91$ async recovery	MLP: $r_{\text{neural}} = 0.881$, stable
Error robustness	4σ detection: $> 99.99\%$ precision	All decoders: $\rho > 0.994$ at $\text{SER} = 10^{-2}$
System integration	Establishes event stream	Traditional decoders safely decode noisy streams

Project 1 answers *how to generate and transmit events efficiently*; Project 2 answers *how to decode those events safely under transmission errors*. Together they suggest that **event-driven acquisition + MLP (or Ridge) decoding** constitutes a practical design principle for low-power

implantable BCIs, offering a favourable trade-off between accuracy and computational cost.

4.2. Future Directions

The performance gap between the best traditional decoder (MLP, $r_{\text{neural}} = 0.881$) and the SNN ($r_{\text{neural}} = 0.932$) suggests that the SNN's advantage lies in its spike-timing-aware feature representation. Future work should explore:

- Burst-aware loss functions and attention mechanisms to recover millisecond-resolution temporal information discarded by 10 ms binning, potentially closing the $\Delta r_{\text{neural}} \approx 0.051$ gap between MLP and SNN.
- Systematic SER-robustness comparison under burst-error wireless channel models (replacing i.i.d. Bernoulli), to validate generalisability of the observed robustness.
- Validation with the actual ASBIT channel model [1].
- Trial-level cross-validation splits to obtain conservative, leak-free performance estimates.

4.3. Study Limitations

- **Project 1:** Synthetic data do not capture in-vivo noise structure or inter-channel correlations; scalability beyond 4 channels is unverified.
- **Project 2:** SER is modelled as i.i.d. Bernoulli, ignoring the burst-error patterns of real wireless channels; results are from a single subject, limiting cross-subject generalisation.
- **Affine correction:** Scale parameters were estimated from test-fold statistics (Eq. 10), which is infeasible in prospective deployment; a separate calibration set should be used in online settings.
- **Cross-validation granularity:** The 5-fold split operates at the bin level, ignoring trial boundaries. This may introduce mild data leakage at trial transitions, slightly inflating reported r_{neural} values.

5. Conclusion

This paper quantifies three key claims for low-power BCI design:

1. An event-driven decoding pipeline maintains $R^2 \geq 0.91$ while reducing CPU active time by **91.35%** relative to 50 ms binning.
2. Among four benchmarked decoders, MLP achieves the highest baseline accuracy ($r_{\text{neural}} = 0.881$), approaching the SNN baseline (0.932). All four traditional decoders maintain SER retention ratios above **0.994** at $\text{SER} = 10^{-2}$, greatly exceeding the SNN (≈ 0.700).
3. The near-perfect SER robustness of all traditional decoders is a *decoder-agnostic* property of rate-coding aggregation, not specific to any single architecture. The combination of event-driven acquisition and traditional decoding simultaneously satisfies the accuracy, energy-efficiency, and error-robustness requirements of implantable wireless BCIs.

Acknowledgements

The author thanks the CRCNS repository maintainers for making the pmd-1 dataset publicly available.

References

- [1] A. Nurmikko et al., “Event-driven wireless neural interface with spike error rate tolerance,” *Nature Electronics*, 2024.
- [2] M. G. Perich et al., “A neural population mechanism for rapid learning,” *Neuron*, vol. 100, no. 4, pp. 964–976, 2018.
- [3] CRCNS pmd-1 Dataset. <https://crcns.org/data-sets/motor-cortex/pmd-1>.
- [4] A. L. Hodgkin and A. F. Huxley, “A quantitative description of membrane current and its application to conduction and excitation in nerve,” *Journal of Physiology*, vol. 117, no. 4, pp. 500–544, 1952.

- [5] R. Q. Quiroga, Z. Nadasdy, and Y. Ben-Shaul, "Unsupervised spike detection and sorting with wavelets and superparamagnetic clustering," *Neural Computation*, vol. 16, no. 8, pp. 1661–1687, 2004.
- [6] A. P. Georgopoulos, A. B. Schwartz, and R. E. Kettner, "Neuronal population coding of movement direction," *Science*, vol. 233, no. 4771, pp. 1416–1419, 1986.
- [7] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.
- [8] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [9] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.